Network Lab Troubleshooting Summary

ISP-BR1 / WAN-BR1 / FortiGate lab

Topology

```
Net/NAT (192.168.85.0/24, gateway .2)
  |-- ISP-HQ  (192.168.85.201)
  '-- ISP-BR1 Gi0/0 (.202) -- Gi0/1 203.0.113.9/30
        '-- WAN-BR1 Gi0/1 (.10) -- Gi0/0 203.0.113.13/30
              '-- FortiGate port1 (203.0.113.14)
                    '-- port2 -> VPC LAN (no IP yet)
```

Problems and fixes

Device	Symptom	Cause	Fix
ISP-BR1	8.8.8.8 failed; .85.2 worked	Two default routes (via .2 and .1). .1 is the VMware host adapter and drops traffic.	no ip route 0.0.0.0 0.0.0.0 192.168.85.1
WAN-BR1	Only 203.0.113.9 pinged	No default route	ip route 0.0.0.0 0.0.0.0 203.0.113.9
FortiGate	Only 203.0.113.13 pinged	No default route	config router static: dst 0.0.0.0/0, gateway 203.0.113.13, device port1
ISP-BR1	Return traffic for 203.0.113.12/30 lost	No route back to WAN-BR1/FortiGate link	ip route 203.0.113.12 255.255.255.252 203.0.113.10
ISP-BR1	Lab networks can't reach internet	NAT gateway doesn't know 203.0.113.x	NAT overload: Gi0/0 ip nat outside, Gi0/1 ip nat inside, ACL 203.0.113.0/24

The pattern behind all of it

• A router only knows its directly connected networks. Anything else needs a route; the default route is the usual answer.

• Routing must work in both directions. If the forward path looks right but ping fails, check the return path.

• The NAT gateway (.2) doesn't know the lab's 203.0.113.x ranges, so the ISP routers must NAT them to the 192.168.85.x address.

Still open

1. FortiGate GUI not reachable (https://203.0.113.14). Check:

- allowaccess on port1 includes https: show system interface port1
- PC has a route to 203.0.113.0/24 via 192.168.85.202: route add 203.0.113.0 mask 255.255.255.0 192.168.85.202
- tracert 203.0.113.14 from the PC to see where it stops

2. port2 LAN not built. Needs an IP, addresses for the VPC, and a firewall policy port2 to port1 with NAT enabled.

Debug checklist

1. show ip route (FortiGate: get router info routing-table all) to see if a route exists
2. Ping the next hop, then the next network, then 8.8.8.8, to find where it breaks
3. Check the return path on every hop
4. On the FortiGate, confirm a firewall policy exists for traffic going through it

5. show ip nat translations on ISP-BR1 to confirm NAT is happening